

Sigrok BMS Protocol Decoder Capture Analysis

Generated from offline sigrok/PulseView analysis outputs

Executive Summary

30

total captures

17

RS485/UART captures

13

CAN captures

19 / 6 / 5

Bridge / Forward / Direct

Key observations

- Anenji Pylon RS485 JKBMS shows the clearest RS485 latency change: Direct cable is -4.58 ms (-47.2%) versus Bridge for average request-to-response latency.
- Growatt RS485 JKBMS has slower worst-case exchange timing: Direct cable is +21.48 ms (+40.1%) versus Bridge for full-exchange P95 latency.
- Growatt CAN SeplosBMS cycles are shorter in Direct cable mode: Direct cable is -164.25 ms (-19.3%) versus Bridge for average CAN cycle duration.
- Growatt CAN JKBMS frame rate stays effectively flat across modes: Direct cable is +0.1 frames/s (+0.2%) versus Bridge.

Capture Context Metadata

This table separates the decoded bus from the physical devices used during the test. Inverter protocol values may come from the captured bus or from fixed test metadata for inverter models with a single tested protocol. **not captured** means the opposite-side protocol is not proven by that capture or by test metadata; it is not inferred from the comparison group name.

Table 1: Capture context metadata.

Mode	Capture	Captured side	Captured protocol	Inverter device	Inverter protocol	BMS device	BMS protocol
Bridge	bridge/china-tower-modbus-rs485	BMS side	China Tower Modbus RS485	Growatt	Pylon CAN	JKBMS	China Tower Modbus RS485
Bridge	bridge/easun-daly-rs485	BMS side	Daly RS485	Easun	Pylon CAN	Daly BMS	Daly RS485
Bridge	bridge/growatt-rs485	Inverter side	Growatt RS485	Growatt	Growatt RS485	JKBMS	not captured
Bridge	bridge/growatt-seplos-rs485	Inverter side	Growatt RS485	Growatt	Growatt RS485	SeplosBMS	not captured
Direct cable	direct/growatt-rs485	Shared bus	Growatt RS485	Growatt	Growatt RS485	JKBMS	Growatt RS485
Bridge	bridge_forward/growatt-rs485	Shared bus	Growatt RS485	Growatt	Growatt RS485	JKBMS	Growatt RS485
Forward	bridge_forward/growatt-seplos-rs485	Shared bus	Growatt RS485	Growatt	Growatt RS485	SeplosBMS	Growatt RS485
Bridge	bridge/jkbms-modbus-rs485	BMS side	JKBMS Modbus RS485	Growatt	not captured	JKBMS	JKBMS Modbus RS485
Bridge	bridge/pace-modbus-rs485	BMS side	PACE Modbus RS485	Growatt	not captured	JKBMS	PACE Modbus RS485
Bridge	bridge/pylon-rs485	Inverter side	Pylon RS485	Growatt	Pylon RS485	JKBMS	not captured
Bridge	bridge/anenji-pylon-rs485	Inverter side	Pylon RS485	Anenji	Pylon RS485	JKBMS	not captured
Direct cable	direct/anenji-jkbms-pylon-rs485	Shared bus	Pylon RS485	Anenji	Pylon RS485	JKBMS	Pylon RS485
Direct cable	direct/anenji-seplos-pylon-rs485	Shared bus	Pylon RS485	Anenji	Pylon RS485	SeplosBMS	Pylon RS485
Bridge	bridge_forward/anenji-pylon-rs485	Shared bus	Pylon RS485	Anenji	Pylon RS485	JKBMS	Pylon RS485
Forward	bridge_forward/anenji-seplos-pylon-rs485	Shared bus	Pylon RS485	Anenji	Pylon RS485	SeplosBMS	Pylon RS485
Bridge	bridge/voltronic-modbus-rs485	BMS side	Voltronic Modbus RS485	Growatt	Pylon RS485	SeplosBMS	Voltronic Modbus RS485
Bridge	bridge/wow-modbus-rs485	BMS side	WOW Modbus RS485	Growatt	Pylon RS485	JKBMS	WOW Modbus RS485
Bridge	bridge/deye-can	Inverter side	Deye CAN	Growatt	Deye CAN	JKBMS	not captured
Bridge	bridge/goodwe-can	Inverter side	GoodWe CAN	Growatt	GoodWe CAN	SeplosBMS	not captured
Bridge	bridge/growatt-can	Inverter side	Growatt CAN	Growatt	Growatt CAN	JKBMS	not captured
Bridge	bridge/growatt-seplos-can	Inverter side	Growatt CAN	Growatt	Growatt CAN	SeplosBMS	not captured
Direct cable	direct/growatt-can	Shared bus	Growatt CAN	Growatt	Growatt CAN	JKBMS	Growatt CAN
Direct cable	direct/growatt-seplos-can	Shared bus	Growatt CAN	Growatt	Growatt CAN	SeplosBMS	Growatt CAN
Bridge	bridge_forward/growatt-can	Shared bus	Growatt CAN	Growatt	Growatt CAN	JKBMS	Growatt CAN
Forward	bridge_forward/growatt-seplos-can	Shared bus	Growatt CAN	Growatt	Growatt CAN	SeplosBMS	Growatt CAN
Bridge	bridge/jkbms-can	BMS side	JKBMS CAN	Growatt	Growatt CAN	JKBMS	JKBMS CAN
Bridge	bridge/pylon-can	BMS side	Pylon CAN	Growatt	Growatt CAN	JKBMS	Pylon CAN
Bridge	bridge/sma-can	BMS side	SMA CAN	Growatt	Growatt CAN	SeplosBMS	SMA CAN
Bridge	bridge/sofar-can	BMS side	Sofar CAN	Growatt	Growatt CAN	SeplosBMS	Sofar CAN

Table 1: Capture context metadata, continued.

Mode	Capture	Captured side	Captured protocol	Inverter device	Inverter protocol	BMS device	BMS protocol
Bridge	bridge/victron-can	BMS side	Victron CAN	Growatt	Growatt CAN	SeplosBMS	Victron CAN

Abbreviations: GW=Growatt, JK=JKBMS, Sep=SeplosBMS, Anj=Anenji, Py=Pylon, IC=inter-cycle.

Table 2: Direct cable versus Bridge values, sorted by absolute percentage difference.

Inv. dev.	Inv. proto.	BMS dev.	BMS proto.	Metric	Direct	Bridge	Diff
GW	GW CAN	JK	GW CAN	Cycles/s	0.2	0.1	+0.1 (+100%)
GW	GW CAN	JK	GW CAN	Cycle min (ms)	4980.23	9980.7	-5000.47 (-50.1%)
GW	GW CAN	JK	GW CAN	Cycle avg (ms)	4980.23	9980.7	-5000.47 (-50.1%)
GW	GW CAN	JK	GW CAN	Cycle max (ms)	4980.23	9980.7	-5000.47 (-50.1%)
GW	GW CAN	JK	GW CAN	Cycle P95 (ms)	4980.23	9980.7	-5000.47 (-50.1%)
GW	GW CAN	Sep	GW CAN	Cycle min (ms)	129.93	251.23	-121.3 (-48.3%)
Anj	Py 485	JK	Py 485	Req->Rsp avg (ms)	5.12	9.71	-4.58 (-47.2%)
GW	GW CAN	Sep	GW CAN	IC gap min (ms)	168.23	114.67	+53.56 (+46.7%)
GW	GW CAN	Sep	GW CAN	Cycle max (ms)	831.77	1496.22	-664.45 (-44.4%)
Anj	Py 485	JK	Py 485	Req->Rsp P95 (ms)	5.69	9.91	-4.22 (-42.6%)
GW	GW 485	JK	GW 485	Full exchange P95 (ms)	75	53.52	+21.48 (+40.1%)
GW	GW CAN	Sep	GW CAN	Cycle P95 (ms)	831.77	1221.32	-389.55 (-31.9%)
GW	GW 485	JK	GW 485	Full exchange avg (ms)	59.34	49.09	+10.24 (+20.9%)
GW	GW 485	JK	GW 485	Complete exchanges/s	2	2.5	-0.5 (-20%)
GW	GW CAN	Sep	GW CAN	Cycles/s	1.2	1	+0.2 (+20%)
GW	GW CAN	Sep	GW CAN	Cycle avg (ms)	687.13	851.37	-164.25 (-19.3%)
GW	GW CAN	Sep	GW CAN	IC gap avg (ms)	168.23	155.49	+12.74 (+8.2%)
GW	GW 485	JK	GW 485	Req->Rsp avg (ms)	5.13	4.76	+0.37 (+7.7%)
Anj	Py 485	JK	Py 485	Full exchange avg (ms)	102.92	111.16	-8.25 (-7.4%)
Anj	Py 485	JK	Py 485	Full exchange P95 (ms)	145.15	153.03	-7.88 (-5.1%)
GW	GW 485	JK	GW 485	Req->Rsp P95 (ms)	5.44	5.29	+0.14 (+2.7%)
GW	GW CAN	Sep	GW CAN	IC gap max (ms)	168.23	165.78	+2.45 (+1.5%)
GW	GW CAN	Sep	GW CAN	IC gap P95 (ms)	168.23	165.78	+2.45 (+1.5%)
GW	GW CAN	JK	GW CAN	Frames/s	52	51.9	+0.1 (+0.2%)
Anj	Py 485	JK	Py 485	Complete exchanges/s	0.8	0.8	0 (0%)
GW	GW CAN	Sep	GW CAN	Frames/s	13	13	0 (0%)

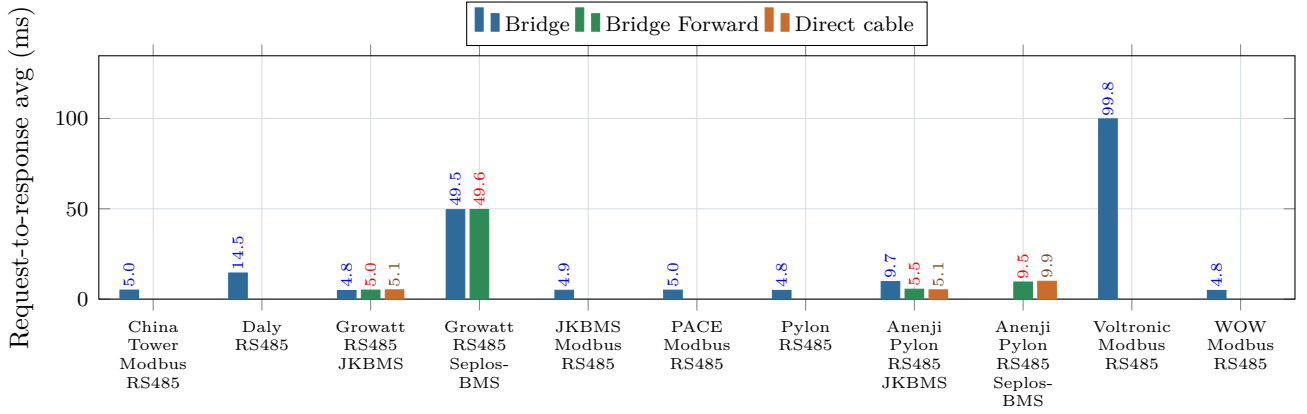
1 Protocol Overview Charts

These charts include every protocol group available in the overview CSV. Missing topology modes are intentionally left blank, so a group can have one, two, or three bars depending on the captures currently available.

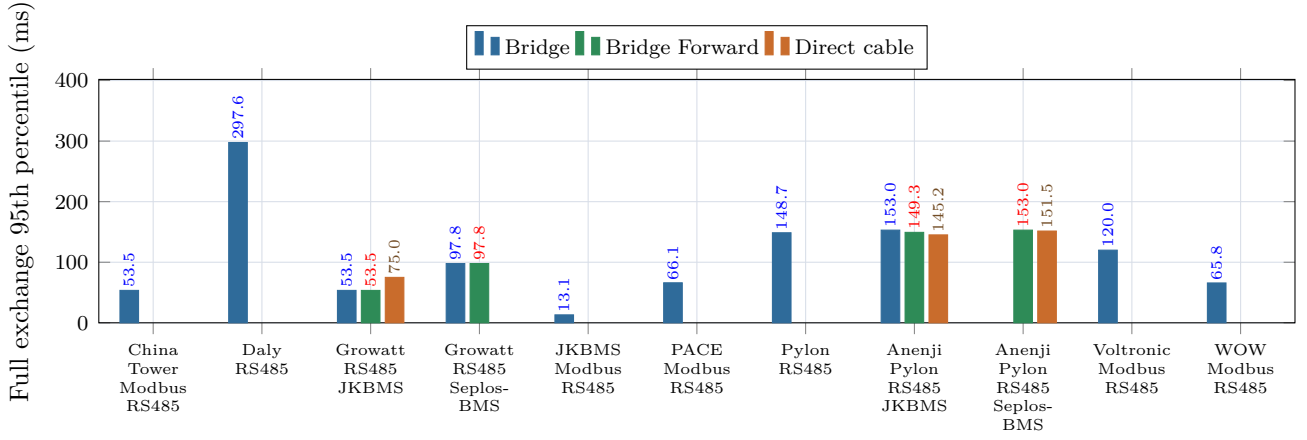
For RS485/UART charts, the timing terms are:

- Request-to-response latency: response start minus request end for a complete paired exchange.
- Full exchange duration: response end minus request start for a complete paired exchange. This includes request transmission, turnaround/processing gap, and response transmission.
- Inter-exchange gap: response end of one complete exchange to request start of the next complete exchange.
- 95th percentile: the value below which 95% of the measured complete request/response pairs fall.

For CAN charts, cycle duration is the time from the first frame in a detected CAN burst to the last frame in that burst. CAN cycle duration charts use a logarithmic y-axis because values span multiple orders of magnitude, from sub-millisecond single-frame cycles to multi-second capture-length cycles. CAN cycle P95 is the closest counterpart to RS485/UART full-exchange P95 in this report.

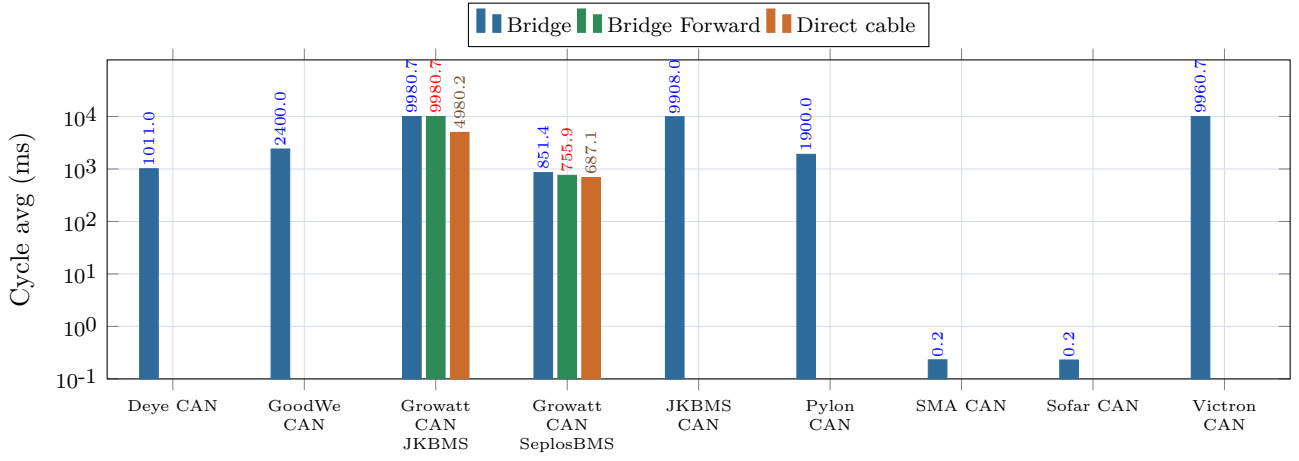


(a) Average request-to-response latency for all available RS485/UART protocol groups.

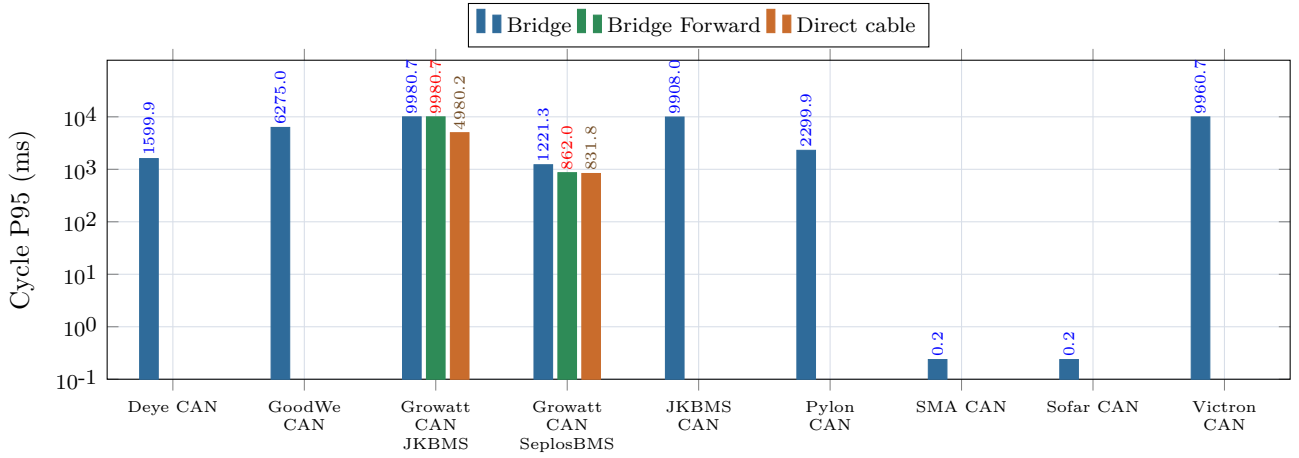


(b) P95 full-exchange duration for all available RS485/UART protocol groups.

Figure 1: RS485/UART protocol overview charts.

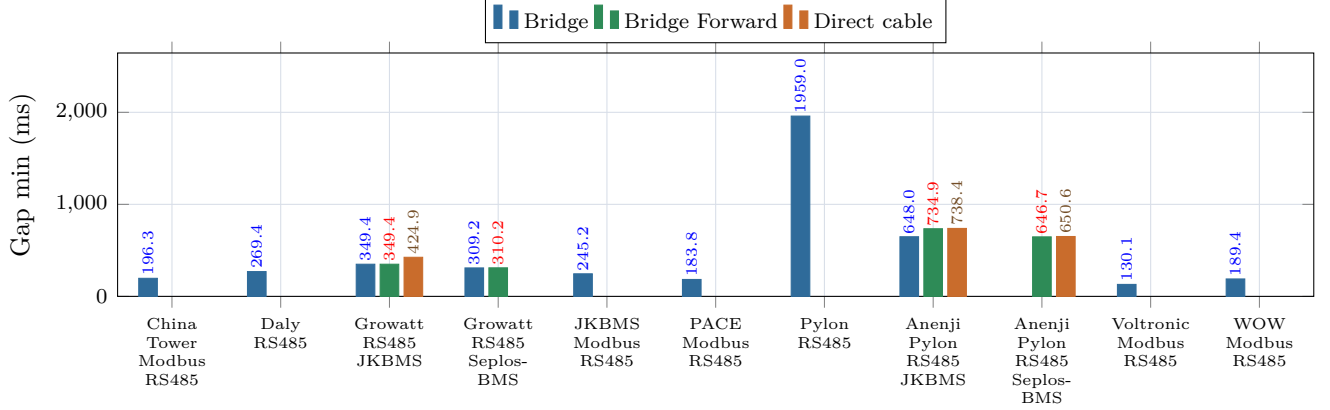


(a) Average CAN cycle duration, log scale.

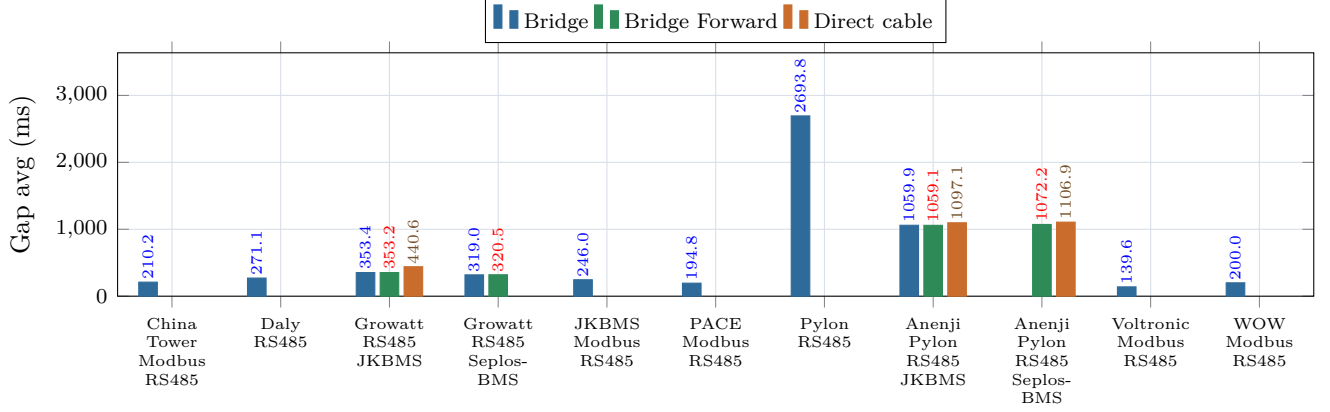


(b) 95th-percentile CAN cycle duration, log scale.

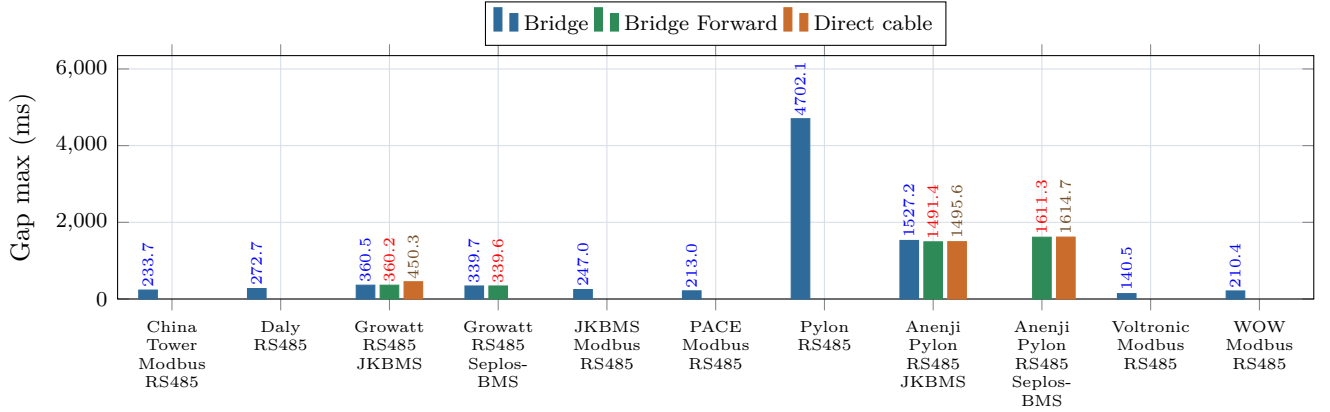
Figure 2: CAN protocol overview charts.



(a) Minimum gap between complete RS485/UART exchanges.

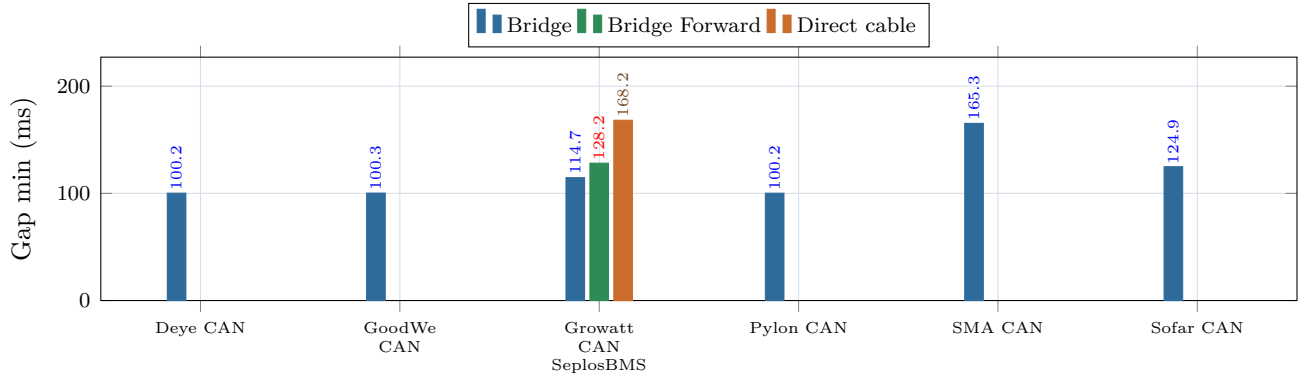


(b) Average gap between complete RS485/UART exchanges.

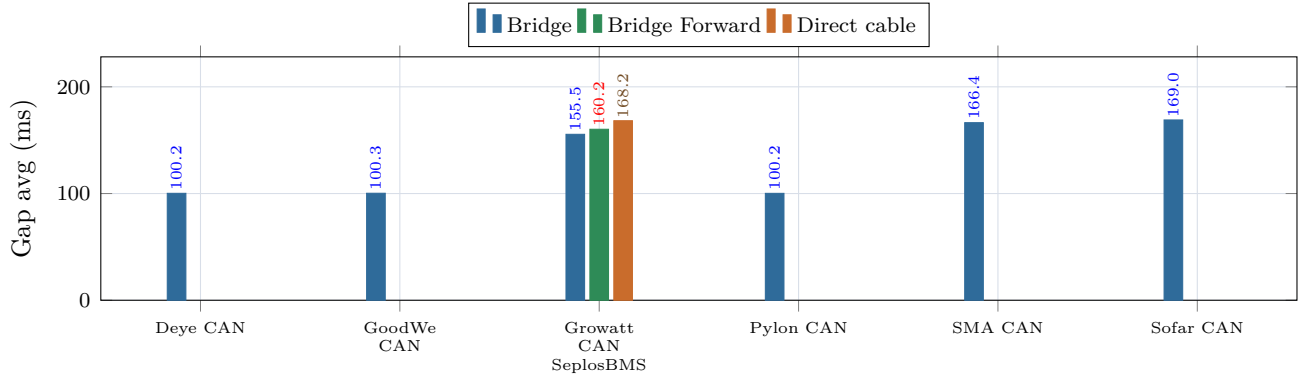


(c) Maximum gap between complete RS485/UART exchanges.

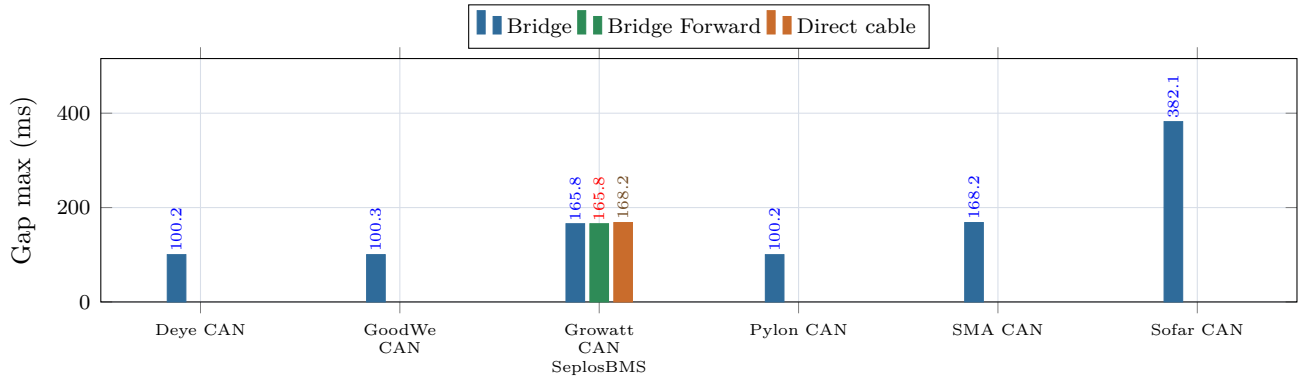
Figure 3: RS485/UART inter-exchange gap overview charts.



(a) Minimum inter-cycle gap for CAN protocol groups with at least two detected cycles.

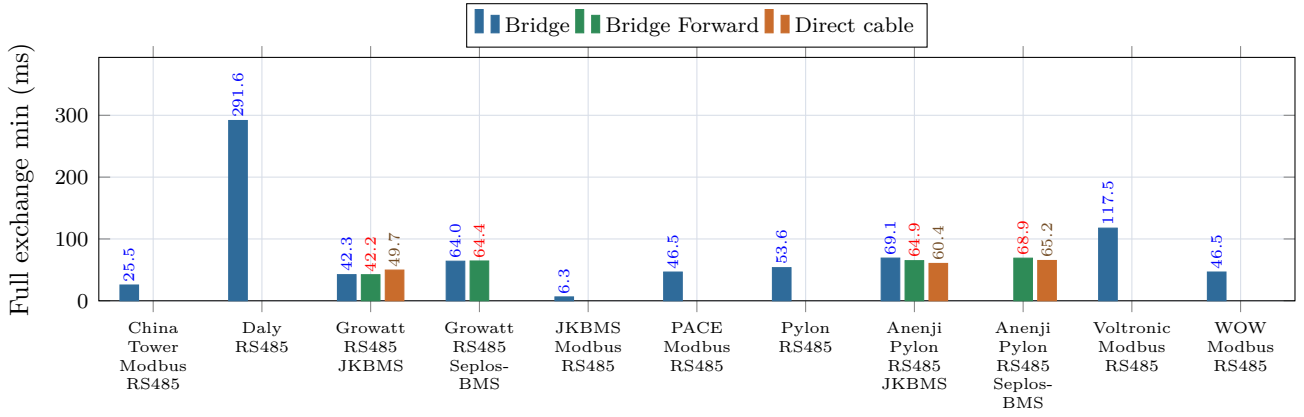


(b) Average inter-cycle gap for CAN protocol groups with at least two detected cycles.

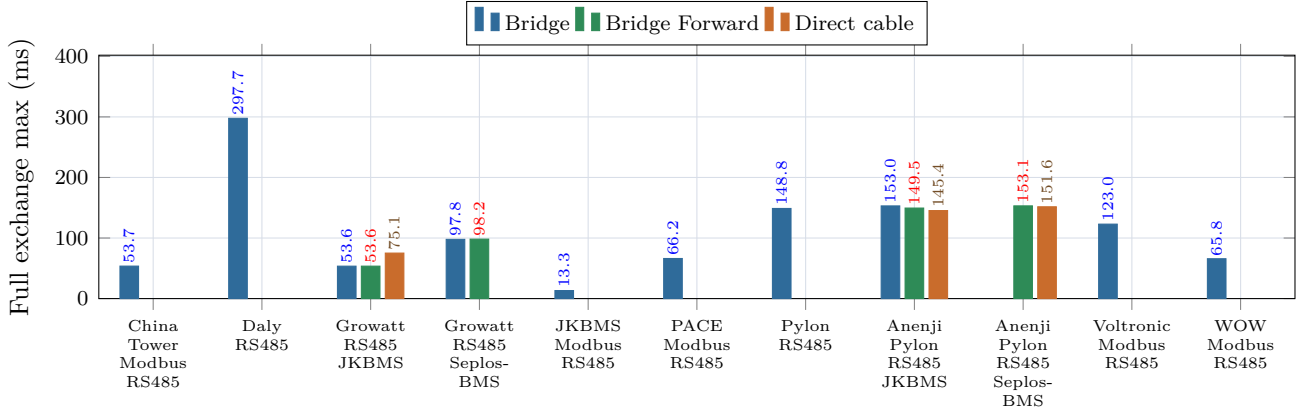


(c) Maximum inter-cycle gap for CAN protocol groups with at least two detected cycles.

Figure 4: CAN inter-cycle gap overview charts.

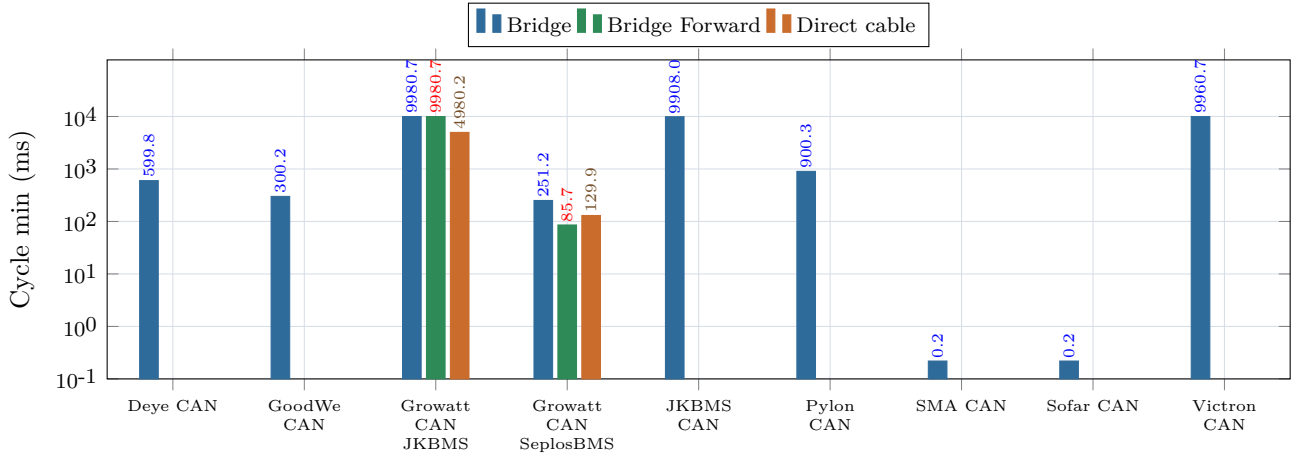


(a) Minimum full-exchange duration for all available RS485/UART protocol groups.

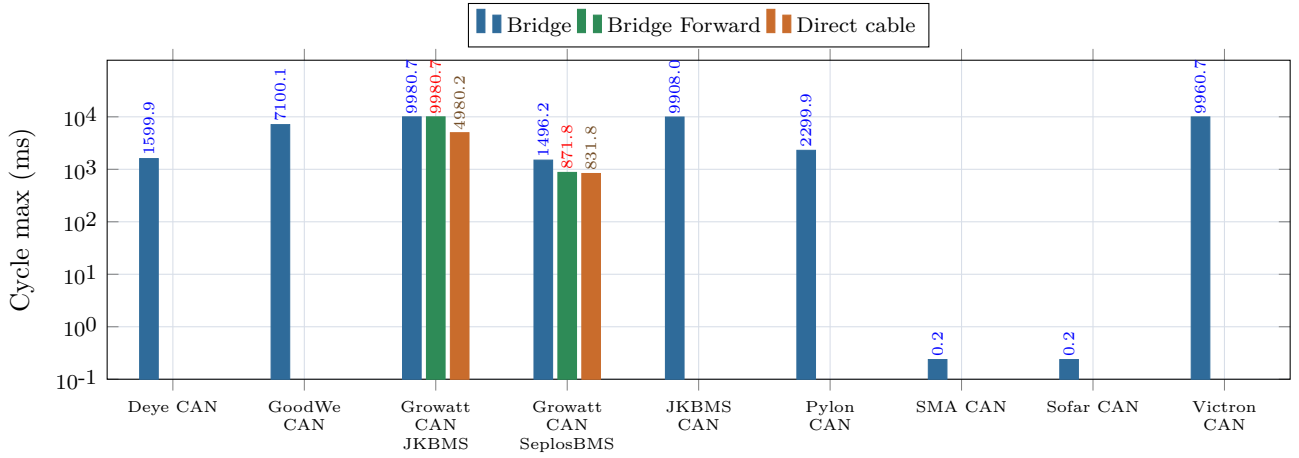


(b) Maximum full-exchange duration for all available RS485/UART protocol groups.

Figure 5: RS485/UART full-exchange range overview charts.



(a) Minimum CAN cycle duration, log scale.



(b) Maximum CAN cycle duration, log scale.

Figure 6: CAN cycle range overview charts.

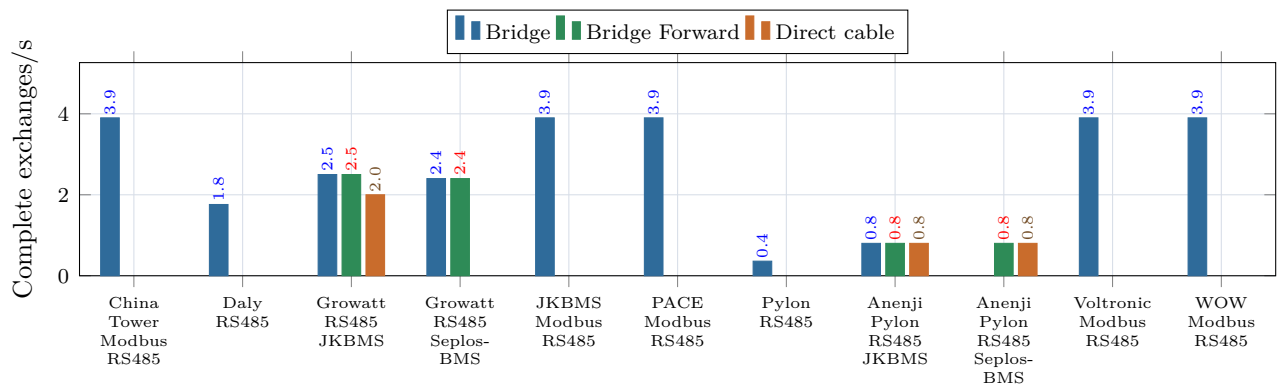


Figure 7: RS485/UART exchange rate overview chart.

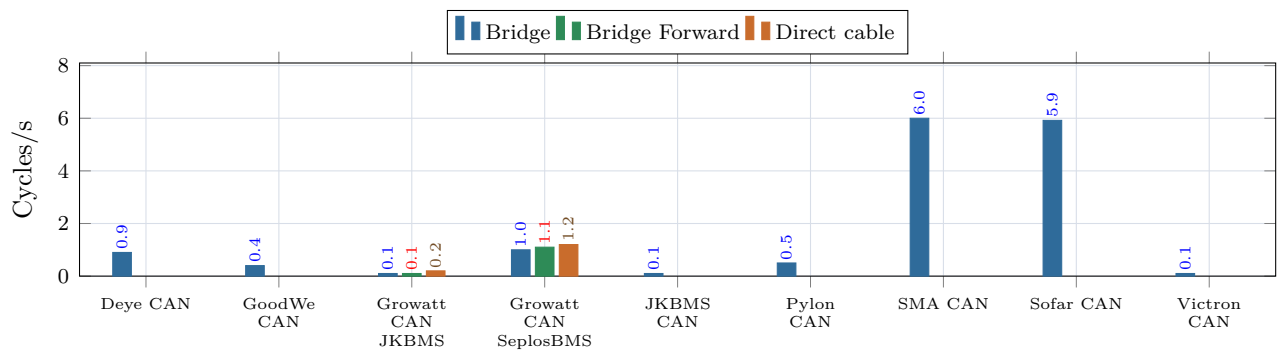


Figure 8: CAN cycle rate overview chart.

2 Three-mode Delta Tables

The following tables focus only on groups where all three modes are available: Bridge, Bridge Forward, and Direct cable. Deltas are directional: orange means an increase, green means a decrease. Timing rows are displayed in milliseconds. Whether an increase is good depends on the metric.

2.1 RS485/UART Three-mode Deltas

Comparison group	Metric	Bridge	Forward	Direct	Forward vs Bridge	Direct vs Bridge	Direct vs Forward
Growatt RS485 JKBMS	Complete exchanges/s	2.5	2.5	2	0 (0%)	-0.5 (-20%)	-0.5 (-20%)
Growatt RS485 JKBMS	Req->Rsp avg (ms)	4.76	5.01	5.13	+0.25 (+5.2%)	+0.37 (+7.7%)	+0.12 (+2.4%)
Growatt RS485 JKBMS	Req->Rsp P95 (ms)	5.29	5.29	5.44	0 (0%)	+0.14 (+2.7%)	+0.14 (+2.7%)
Growatt RS485 JKBMS	Full exchange avg (ms)	49.09	48.93	59.34	-0.17 (-0.3%)	+10.24 (+20.9%)	+10.41 (+21.3%)
Growatt RS485 JKBMS	Full exchange P95 (ms)	53.52	53.48	75	-0.04 (-0.1%)	+21.48 (+40.1%)	+21.52 (+40.2%)
Anenji Pylon RS485 JKBMS	Complete exchanges/s	0.8	0.8	0.8	0 (0%)	0 (0%)	0 (0%)
Anenji Pylon RS485 JKBMS	Req->Rsp avg (ms)	9.71	5.49	5.12	-4.22 (-43.5%)	-4.58 (-47.2%)	-0.36 (-6.6%)
Anenji Pylon RS485 JKBMS	Req->Rsp P95 (ms)	9.91	6.19	5.69	-3.72 (-37.6%)	-4.22 (-42.6%)	-0.5 (-8%)
Anenji Pylon RS485 JKBMS	Full exchange avg (ms)	111.16	106.95	102.92	-4.21 (-3.8%)	-8.25 (-7.4%)	-4.03 (-3.8%)
Anenji Pylon RS485 JKBMS	Full exchange P95 (ms)	153.03	149.32	145.15	-3.71 (-2.4%)	-7.88 (-5.1%)	-4.17 (-2.8%)

2.2 CAN Three-mode Deltas

Comparison group	Metric	Bridge	Forward	Direct	Forward vs Bridge	Direct vs Bridge	Direct vs Forward
Growatt CAN JKBMS	Frames/s	51.9	52	52	+0.1 (+0.2%)	+0.1 (+0.2%)	0 (0%)
Growatt CAN JKBMS	Cycles/s	0.1	0.1	0.2	0 (0%)	+0.1 (+100%)	+0.1 (+100%)
Growatt CAN JKBMS	Cycle min (ms)	9980.7	9980.68	4980.23	-0.02 (0%)	-5000.47 (-50.1%)	-5000.45 (-50.1%)
Growatt CAN JKBMS	Cycle avg (ms)	9980.7	9980.68	4980.23	-0.02 (0%)	-5000.47 (-50.1%)	-5000.45 (-50.1%)
Growatt CAN JKBMS	Cycle max (ms)	9980.7	9980.68	4980.23	-0.02 (0%)	-5000.47 (-50.1%)	-5000.45 (-50.1%)
Growatt CAN JKBMS	Cycle P95 (ms)	9980.7	9980.68	4980.23	-0.02 (0%)	-5000.47 (-50.1%)	-5000.45 (-50.1%)
Growatt CAN SeplosBMS	Frames/s	13	12.9	13	-0.1 (-0.8%)	0 (0%)	+0.1 (+0.8%)
Growatt CAN SeplosBMS	Cycles/s	1	1.1	1.2	+0.1 (+10%)	+0.2 (+20%)	+0.1 (+9.1%)
Growatt CAN SeplosBMS	Cycle min (ms)	251.23	85.69	129.93	-165.54 (-65.9%)	-121.3 (-48.3%)	+44.24 (+51.6%)
Growatt CAN SeplosBMS	Cycle avg (ms)	851.37	755.94	687.13	-95.43 (-11.2%)	-164.25 (-19.3%)	-68.82 (-9.1%)
Growatt CAN SeplosBMS	Cycle max (ms)	1496.22	871.81	831.77	-624.41 (-41.7%)	-664.45 (-44.4%)	-40.05 (-4.6%)
Growatt CAN SeplosBMS	Cycle P95 (ms)	1221.32	861.95	831.77	-359.37 (-29.4%)	-389.55 (-31.9%)	-30.18 (-3.5%)
Growatt CAN SeplosBMS	Inter-cycle gap min (ms)	114.67	128.17	168.23	+13.51 (+11.8%)	+53.56 (+46.7%)	+40.05 (+31.2%)
Growatt CAN SeplosBMS	Inter-cycle gap avg (ms)	155.49	160.23	168.23	+4.73 (+3%)	+12.74 (+8.2%)	+8 (+5%)
Growatt CAN SeplosBMS	Inter-cycle gap max (ms)	165.78	165.78	168.23	0 (0%)	+2.45 (+1.5%)	+2.45 (+1.5%)
Growatt CAN SeplosBMS	Inter-cycle gap P95 (ms)	165.78	165.78	168.23	0 (0%)	+2.45 (+1.5%)	+2.45 (+1.5%)

3 Capture Overview

3.1 RS485/UART Capture Overview

Captured protocol	Topology	Frames	Loss rate	Req->Rsp avg (ms)	Req->Rsp P95 (ms)	Full P95 (ms)
China Tower Modbus RS485	Bridge	79	2.5%	5.02	5.34	53.53
Daly RS485	Bridge	88	0%	14.46	18.06	297.65
Growatt RS485	Bridge	50	0%	4.76	5.29	53.52
Growatt RS485	Bridge	49	4%	49.52	49.86	97.81
Growatt RS485	Direct cable	100	0%	5.13	5.44	75
Growatt RS485	Bridge Forward	50	0%	5.01	5.29	53.48
Growatt RS485	Bridge Forward	49	4%	49.6	50.49	97.85
JKBMS Modbus RS485	Bridge	78	0%	4.9	5.19	13.12
PACE Modbus RS485	Bridge	79	2.5%	4.99	5.26	66.06
Pylon RS485	Bridge	49	21.74%	4.82	5.54	148.67
Pylon RS485	Bridge	20	33.33%	9.71	9.91	153.03
Pylon RS485	Direct cable	61	51.22%	5.12	5.69	145.15
Pylon RS485	Direct cable	50	33.33%	9.89	12.04	151.5
Pylon RS485	Bridge Forward	24	50%	5.49	6.19	149.32
Pylon RS485	Bridge Forward	20	33.33%	9.47	9.88	153
Voltronic Modbus RS485	Bridge	78	0%	99.8	101.84	119.96
WOW Modbus RS485	Bridge	78	0%	4.8	5.09	65.81

3.2 CAN Capture Overview

Captured protocol	Topology	Frames	IDs	Cycles	Cycle min (ms)	Cycle avg (ms)	Cycle max (ms)	Cycle P95 (ms)	Errors
Deye CAN	Bridge	100	8	9	599.85	1011.01	1599.87	1599.87	0
GoodWe CAN	Bridge	100	8	4	300.25	2399.97	7100.06	6275.02	0
Growatt CAN	Bridge	519	10	1	9980.7	9980.7	9980.7	9980.7	0
Growatt CAN	Bridge	130	12	10	251.23	851.37	1496.22	1221.32	0
Growatt CAN	Direct cable	260	10	1	4980.23	4980.23	4980.23	4980.23	0
Growatt CAN	Direct cable	65	12	6	129.93	687.13	831.77	831.77	0
Growatt CAN	Bridge Forward	520	10	1	9980.68	9980.68	9980.68	9980.68	0
Growatt CAN	Bridge Forward	129	12	11	85.69	755.94	871.81	861.95	0
JKBMS CAN	Bridge	344	12	1	9907.99	9907.99	9907.99	9907.99	0
Pylon CAN	Bridge	100	8	5	900.28	1900.01	2299.95	2299.95	0
SMA CAN	Bridge	60	6	60	0.22	0.23	0.24	0.24	0
Sofar CAN	Bridge	148	6	148	0.22	0.23	0.24	0.24	0
Victron CAN	Bridge	250	19	1	9960.73	9960.73	9960.73	9960.73	0

Notes

- The LaTeX source is generated from committed CSV files under `analysis/out/`.
- Capture durations are not identical, so normalized rates are included where useful.
- CAN cycle duration can be dominated by capture length when only one cycle is detected.
- Compiled PDFs and LaTeX build artifacts should stay out of the repository.